



Architecting Veeam Backup for Microsoft365 v8 At Scale



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Agenda

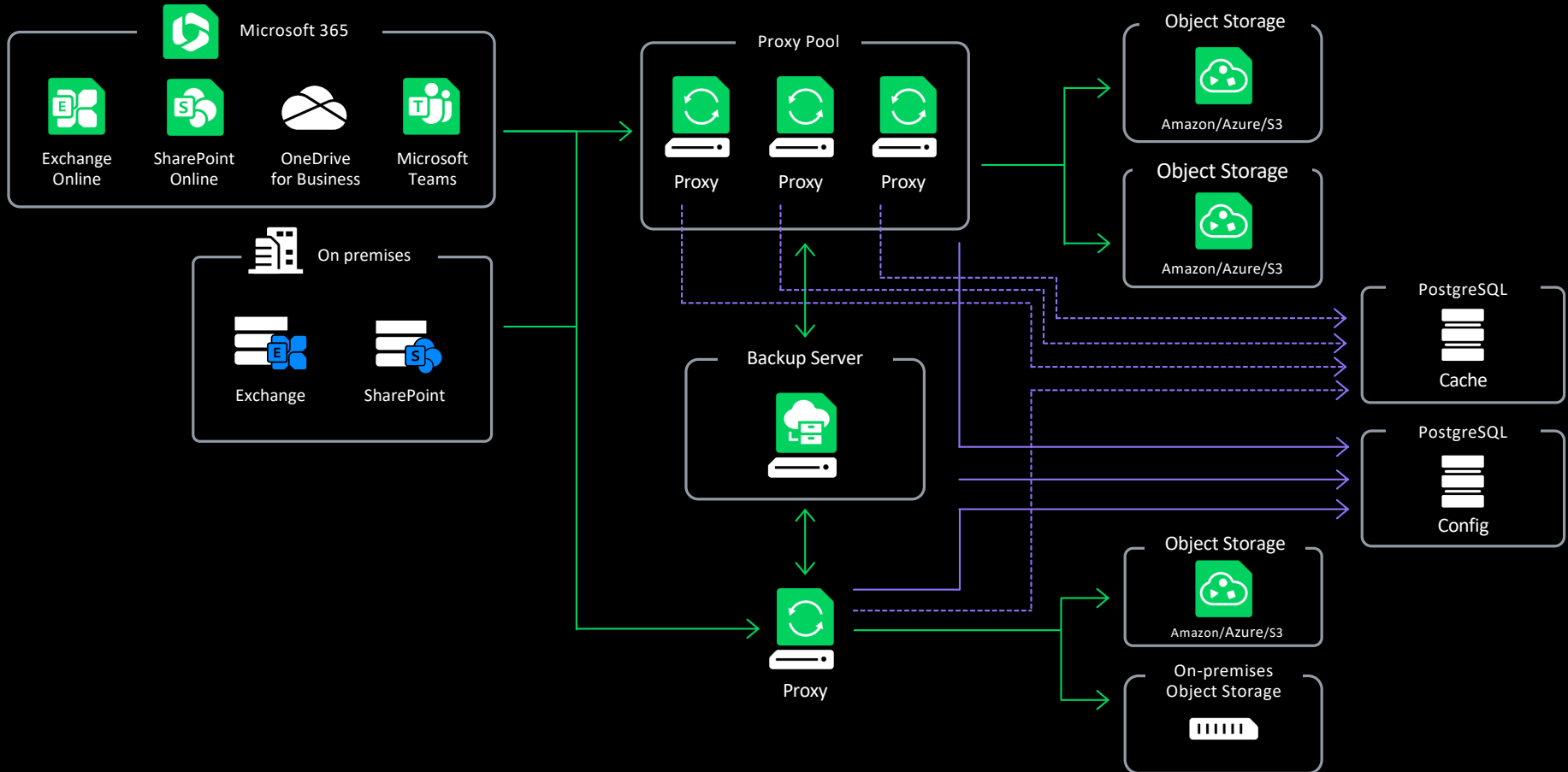
1. VB365 v7 Recap
2. Scaling & Monitoring Infrastructure
3. Backup Job Optimization

Where We Were in v7



- Tightly coupled relationships
job: repository: proxy
- 2500-4000 objects per proxy max
- Protecting Sharepoint items very slow
- Reassignment of repositories not practical
- Any maintenance (patching, upgrades) painful
- Immutability only on the secondary copy (new tech!)

New v8 architecture



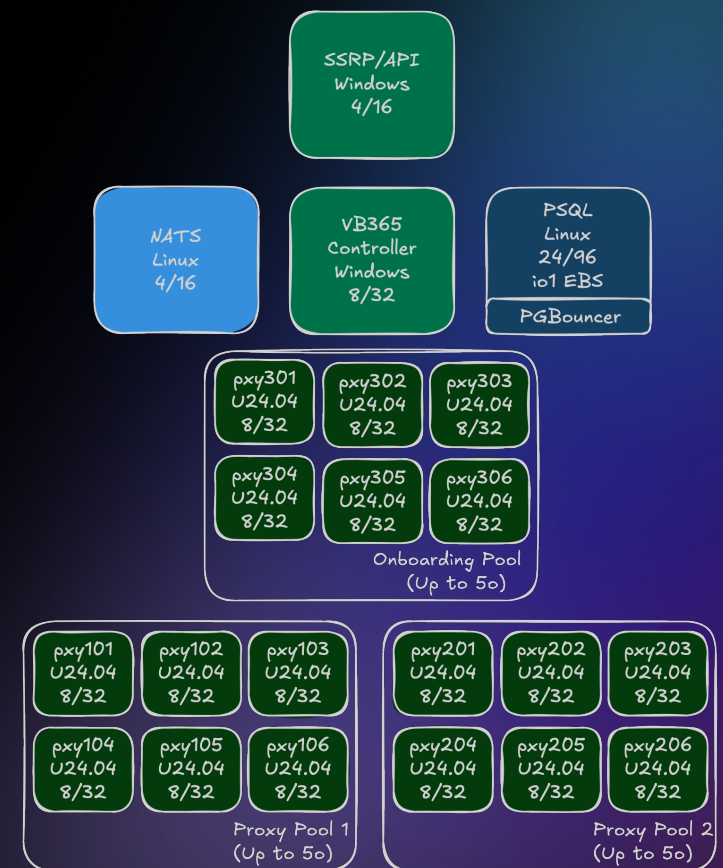
Designing for Scalability

- No all-in-one if you are big enough to be here
- Fully distributed architecture
- When possible, use Linux
- Cloud Always, compute close to storage
- Predict growth, both scale up and scale out
- Database, object storage compromise
- Address Exchange, Sharepoint components separately



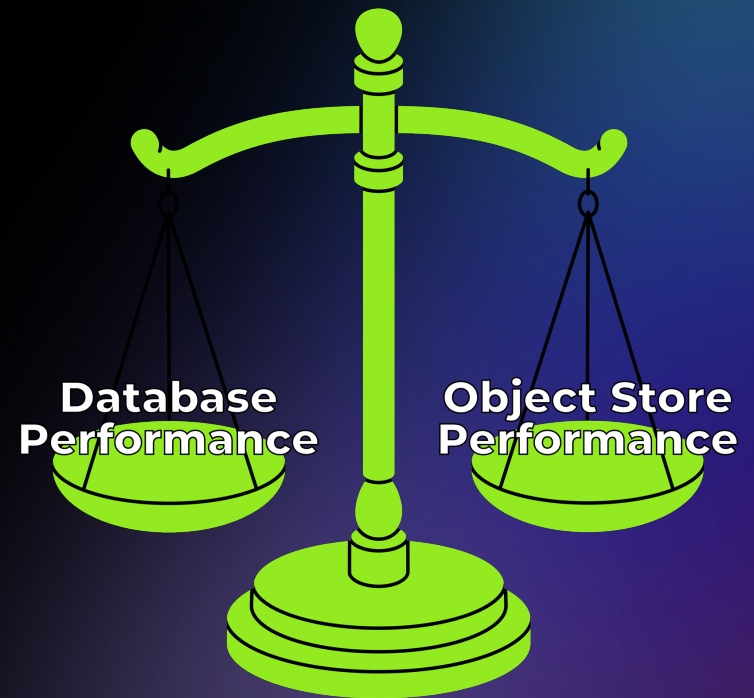
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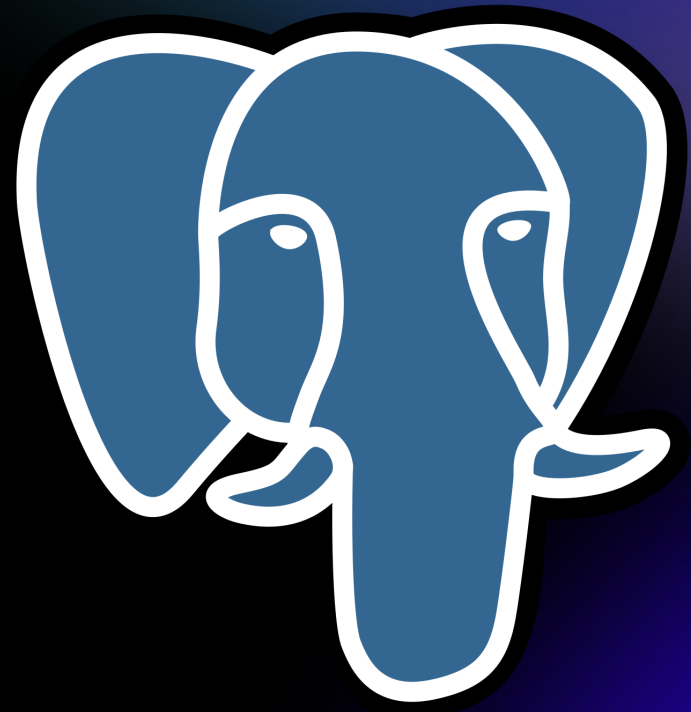
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PostgreSQL

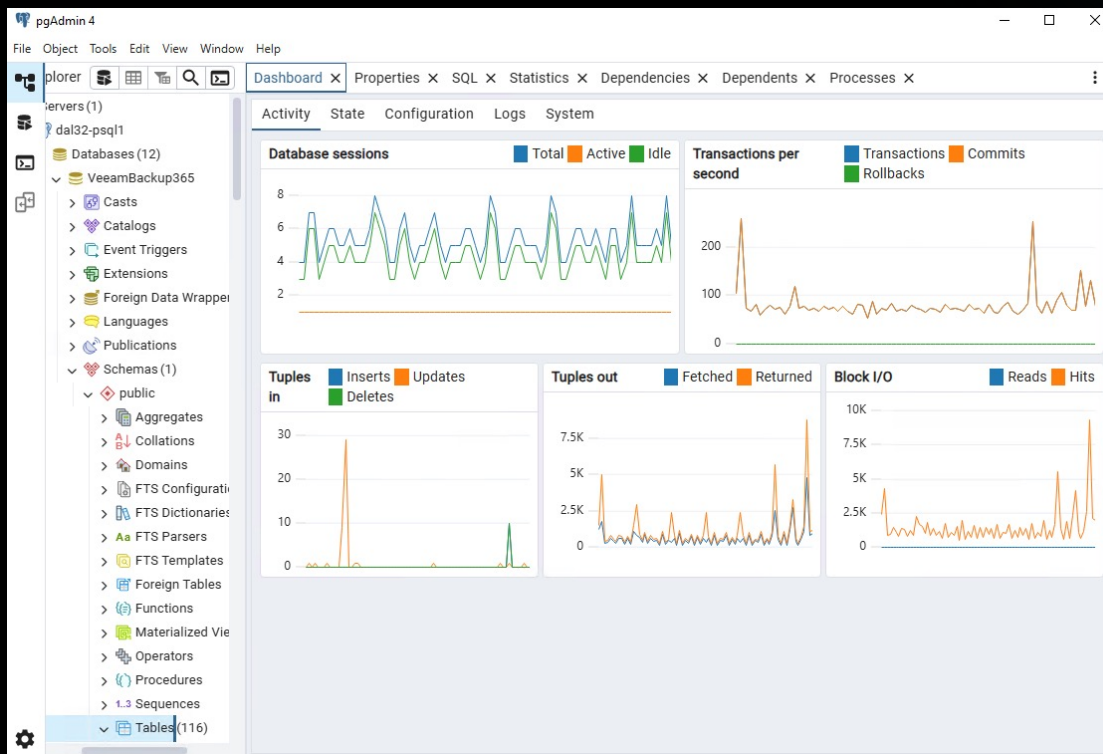
Hardware and Design

- Always external, always Linux
- Hardware (50 proxies): 16 CPU / 64 GB RAM
- Hardware (150 proxies): 24 CPU / 96 GB RAM
- But proxies not the true metric
- Run **Set-VBOPSQLDatabaseLimits** after install & as part of upgrade procedures
- Maximum version: 15.9 or 16.4
- PGBouncer for connection pooling
- Do not share with VBR (VCSP VEX exception)



PostgreSQL

Monitoring



- OS Monitoring
 - RAM
 - CPU
 - Disk I/O
- Use pgadmin4
 - Pg_stat_activity – track connection states
 - 1-2 MB RAM/connection
 - Each proxy up to 100 connections

NATS

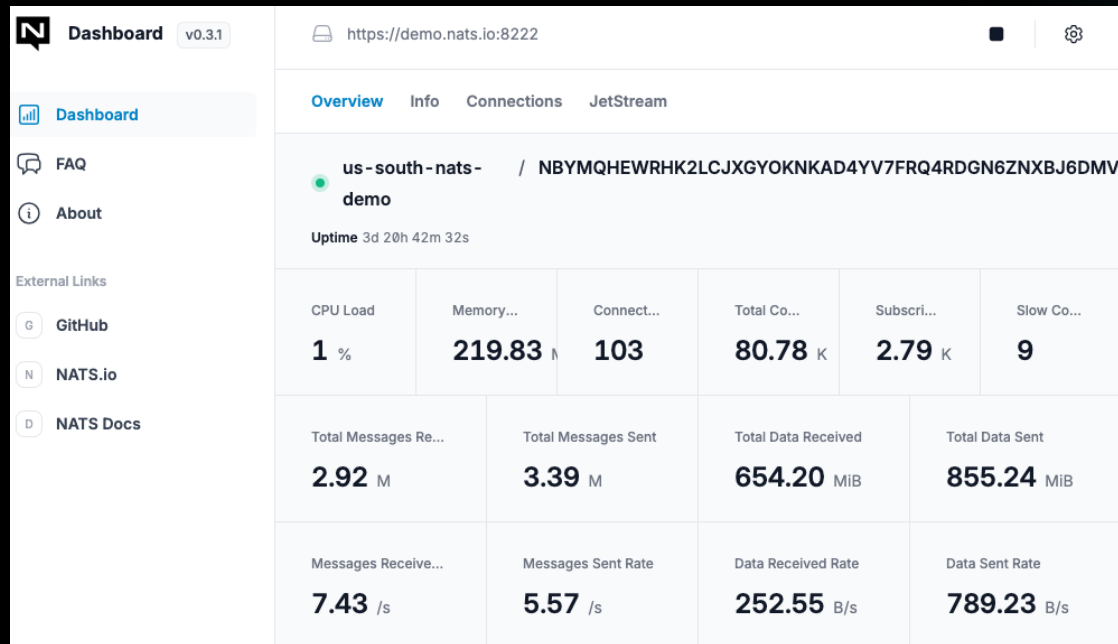
Hardware and Design

- Always External
- Hardware: 8 CPU / 32 GB RAM
- Single node for speed
- Consider containerizing for management
 - How To: <https://tinyurl.com/NatsOnPodman>
- Client Advertise if shared container host
- GOMEMLIMIT – is it right for me?



NATS

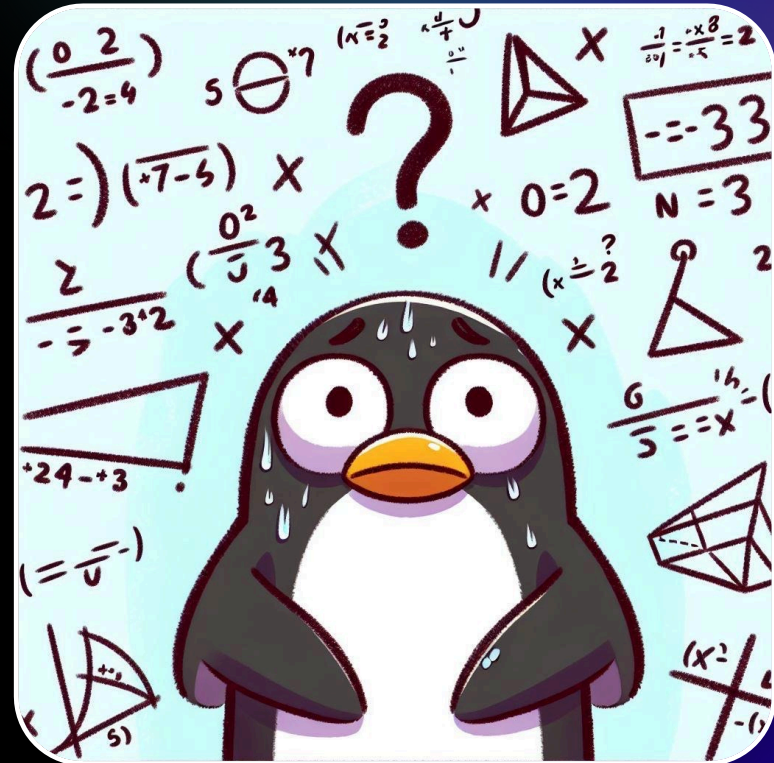
Monitoring



- Tooling
 - NATS Surveyor
 - NATS-Dashboard
- OS Monitoring
 - RAM
 - CPU
 - Disk I/O
- Troubleshooting
 - Clear folder OR
 - Restart

Proxies

- Keep all components close
- Now supports Ubuntu 24.04, RHEL 9.4 in 8.1
- Cloud Instances
 - Azure D or F type instances
 - AWS M series instances
- 1:4 CPU:RAM ratio
 - 4 4x16 proxies more effective than 2 8x32
- No more thread management!



Pools



- Up to 3 pools, 50 proxies per pool
 - Past 50 dedicated management proxy
- Dedicate 1 pool to onboarding orgs
- Start|Get|Stop-VBORepositoryOwnerChangeSession
- Build in mass but cost optimize by powering down

Backup Job Optimization

Discovery

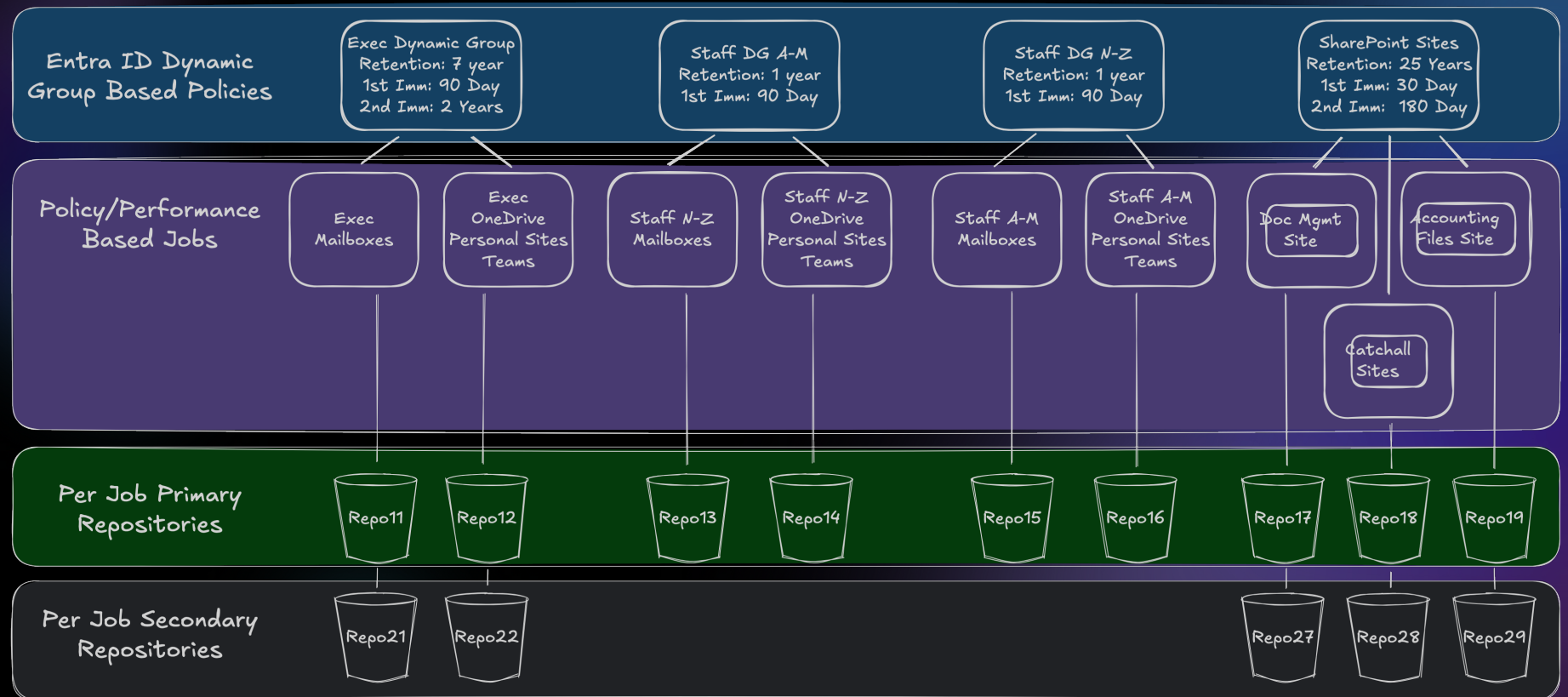
- Gathering Using Data
 - Microsoft Admin Center Usage Reports
 - <https://calculator.veeam.com/vb365>
- Be Aware of
 - Multi-geo organizations
 - Large Sharepoint sites
 - Compliance based retention data units
 - Unlicensed users/groups (shared, group, etc.)
- Design for very conservative splitting, organizations will grow/change

Implementation

- Split Organizations by
 - Retention Policies
 - Exchange / Sharepoint
 - 5000 users
- Add 2nd App Registration for heavy Sharepoint
- Segregate large Sites

Backup Job Optimization

In Practice



Automation

Make your job easier

SharePoint Teams jobs

- Too many sites for a single job?
- Need to make sure Teams process with matching site?
- Don't want to manually remove deleted jobs?

Scripts:

VB365-Job-Manager

- Creates jobs with X sites per job
- Matches Team and SP site into same job
- Can target multiple repos

VB365-Remove-Deleted-SP-Sites-from-jobs

- Checks previous job log for "404 failed to process site"
- Removes site from Job
- Disables & renames job if last site in job is removed
- All logged

VBO-ProxyAutomation

- Polls for running job sessions through a scheduled task
- Automatically stops and starts proxies based on need
- Supports VMware, Azure, AWS

<https://github.com/veeamhub>

Restores

We aren't keeping backups just for fun

Explorers

- Advanced restore capabilities
- Advanced e-discovery like searching
- Data flow: repo > proxy > vb365 > explorer > destination

Powershell

- Uses each Explorer's Powershell commands
- Creates a restore session on VB365 server
- Data flow: repo > proxy > vb365 > explorer > destination

API

- Advanced restore capabilities
- VB365 server or separate machine
- Data flow: repo > proxy > M365
- Data flow: repo > proxy > vb365 > other destinations
- Swagger UI for ease of testing

Restore Portal

- RBAC capabilities / Self Service
- Simplified restores to M365 only
- Data flow: repo > proxy > M365

SaaS FTW!

When In Doubt, Get An Expert

Managed Service



Veeam Cloud & Service Provider (VCSP) Program

- Diverse marketplace of providers and services to provide a tailored solution
- Unlimited, immutable storage available
- Offload expertise to a Veeam Service Provider

NEW! Backup Service



Veeam Data Cloud *for Microsoft 365*

- When first party is a hard requirement
- Infrastructure & updates managed by Veeam
- Unlimited storage included

Q&A

Slides and Materials Available At:

<https://tinyurl.com/Architecting-VB365-v8>



Thank You.

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